UNIVERSITATEA DIN BUCUREȘTI
FACULTATEA DE FILOSOFIE

Lucrare de disertație

From Local to Portable Rigidity

Demonstrative Reference and the Cognitive Ground of Rigid Designation

Coordonator științific:
Prof. Dr. Mircea Dumitru

Absolvent:
Florin Cojocariu

București
2026-05-24

Contents

Abstract

The dissertation argues that Kripke's account of rigid designation rests on a cognitive precondition his semantics presupposes but does not theorize. In Kripke's natural-language gloss of $\forall x \forall y\,(x = y \to \Box(x = y))$ — "for every object x and object y" — the word "object" spans three distinct things: mathematical variables rigid by assignment, encountered instances rigid within an encounter, and named individuals rigid across absence. The formula's validity holds for the first; the philosophical claim concerns the third. Kripke argues for that claim independently and is aware the formal result concerns variables, not names; what he does not provide is a cognitive or metasemantic account of how ordinary speakers achieve the referential stability his semantics attributes to names. That account is what this dissertation supplies.

The thesis is that rigid name use can be illuminated metasemantically by comparison with x^o-mode tracking across absence, where x^o marks the use of a word in coordination with an experienced instance $\{X\}$ and x^c marks its use within broader linguistic relations. The proposal distinguishes two levels of linguistic-cognitive operation: at x^o, a speaker's use is anchored to a particular through live coordination; at x^c, terms function within broader inferential and testimonial networks, with no experienced instance that it refers to. The dissertation argues that the modal force characteristic of rigid designation depends on the former, while the epistemic work of identification and convergence is largely carried out by the latter.

The argument runs through Russell, Kripke, and the post-Kripkean cognitive literature. Russell identified the mechanism — demonstrative reference under acquaintance — and then restricted it to immediate sense-data on Cartesian grounds. Kripke extended the scope of rigidity to natural-language names but inherited Russell's restraint by leaving the mechanism unspecified. Existing cognitive and metasemantic accounts — including Evans's discriminating conception, Recanati's mental files, and Devitt's causal-historical picture — each capture part of the explanatory burden. My claim is not that they ignore the problem, but that they do not yet explain rigid designation in terms of a single primitive coordination that scales from demonstrative encounter to socially transmitted name use. The dissertation locates the mechanism Russell discovered and Kripke extended in x^o-mode tracking, with baptismal lifting carrying that tracking across absence through social practice.

1. Introduction

1.1. The Problem Kripke Left Open

Saul Kripke's modal, epistemic, and semantic arguments in *Naming and Necessity* and "Identity and Necessity" establish that proper names are rigid designators: a name picks out the same individual in every possible world in which that individual exists. The arguments are compelling, and the conclusion is by now a standard starting point in the philosophy of language. What the arguments establish is what rigid designators *do*. What they do not establish, and what Kripke himself does not undertake to establish, is what cognitive operation makes the doing possible.

The gap is specifically metasemantic. A formal apparatus can stipulate that a term holds its referent fixed across worlds. The stipulation can be written down and its logical consequences can be worked out. But stipulation is not explanation. When Kripke writes that reference is fixed by an "initial baptism" and transmitted through a "chain of communication," he describes the practice in which rigidity lives without describing the cognitive operation that makes the practice possible. A baptismal act coordinates a word with an individual; a chain of communication preserves the coordination. Neither is a logical relation. Both are things that cognitive systems do.

The gap is not merely an accident of exposition. Across several major twentieth-century approaches, one finds pressure to let formal or descriptive resources shoulder more of the explanatory burden than they can plausibly bear. Russell, Quine, and Lewis differ profoundly in aim and method, and I do not claim that they share one theory of reference. The narrower claim is diagnostic: each, in different ways, leaves underdescribed the cognitive-practical relation by which speakers get a grip on particular individuals. Kripke's own account is the counterweight to this pattern — his appeal to baptism and causal chains is precisely the recognition that reference to a particular cannot be secured by formal operations alone. What Kripke leaves unanalyzed is the cognitive operation that baptism performs and the mechanism by which its result is preserved once the baptismal community is gone. The dissertation takes up this analysis.

This matters because reference to a particular is not generated by linguistic machinery operating on itself. No tightening of the network of words produces grip on an individual. Descriptions can locate an object of which one has no independent grip, but only by specifying conditions that something in the world happens to satisfy — and what they locate is whoever satisfies the conditions, not the individual as such. Grip on an individual comes from a different direction: from coordination with experience, from using a word at the point where it meets an encounter. Kripke's semantics inherits whatever this coordination accomplishes. The dissertation's question is what the coordination consists in and how the semantics depends on it.

1.2. Two Operations of Fixing

The dissertation distinguishes, from the outset, two operations by which a linguistic element can be tied to a value. They are different operations that do different work, and the distinction between them is the backbone of the positive account developed in Chapters 4 and 5.

The first operation is value-fixing through structures internal to the linguistic system. A variable under assignment is tied to a domain element by the assignment function. A definite description picks out whoever satisfies its conditions. A logical specification singles out what meets its constraints. A mathematical construction yields a specific object through the structural relations that define it. In each case, the tying is completed within the linguistic network — through descriptive content, through the formal apparatus of quantification, through structural specification. The dissertation will call this *A-grounding*, following the analogy with assignment in model theory, though the operation is broader than assignment in the strict sense. A-grounding is the operation that does not reach outside the linguistic system.

The second operation is value-fixing through coordination with something outside the linguistic system. When a speaker looks at a chair and uses the word "chair" to pick out that particular chair, the tying is not achieved through descriptive content or formal specification. It is achieved through a cognitive coordination: the word meets the present object through perceptual engagement, and the meeting itself constitutes the tying. The dissertation will call this *R-grounding*, where the R indicates referential, in the sense of reaching out to something. R-grounding requires the presence or perceptual availability of what is being referred to. It is the operation that reaches outside.

The two operations are not rivals. Each does work the other cannot do. A-grounding is indispensable for formal reasoning, for mathematics, for descriptions when they function descriptively, for talk that does not pretend to reach any particular. R-grounding is indispensable for reference to individuals in the world. The tradition often treats A-grounding as sufficient for both jobs; Kripke is the signal exception, having seen that reference to particulars requires something formal operations cannot supply. What Kripke did not do, and what the dissertation undertakes, is to describe the non-formal operation at the cognitive level and to follow its consequences through the transition from live encounter to later absence. In the vocabulary of Chapter 4, the difference is one of argument type for a single coordination operator R: A-grounding is coordination that stays within the linguistic pattern $<x>$ — the $\mathcal{R}(\mathcal{D}, N^c)$ type, description coordinated with name — while R-grounding is coordination that crosses into experience, the $\mathcal{R}(x^o, E(\{X\}))$ type that reaches a particular through encounter.

Chapter 4 develops both operations in the object-mode/concept-mode vocabulary; for this introduction, what matters is that the two are non-interchangeable.

1.3. The Thesis

The dissertation advances a single thesis. Rigid designation in the Kripkean sense can be fruitfully understood by starting from a more primitive phenomenon — *local rigidity* — present whenever a word coordinates with an experienced instance through R-grounding. When I track an object with my senses and use a word to pick it out, the coordination is rigid: the word picks out that individual without descriptive mediation, and modal claims about it carry no scope ambiguity. There is nothing for scope to interact with; the word just points, and what it points at is what the modal claim is about. Baptism introduces a new linguistic device whose later use preserves, under conditions of absence, the kind of holding-fixed first visible in demonstrative encounter.

The thesis has two parts. The first concerns the origin of rigidity: local rigidity is the primitive phenomenon, produced by live R-grounding in demonstrative encounter. The second concerns how rigidity persists once the original encounter is over. Here the account has to handle a fact that has been underdescribed in the literature: R-grounding requires active coordination with a present particular, and after the particular is gone — after the chair leaves, after Aristotle dies — R is no longer available. Contemporary speakers who use the name "Aristotle" cannot be performing R-grounding with Aristotle; they have no access to him. Some other operation must be at work.

Once a name has moved beyond the baptismal setting, later speakers typically rely on descriptive and testimonial material in using it. On the present view, however, that material mediates the speaker's epistemic access to an already inherited name-use rather than constituting the referent. The account is therefore anti-descriptivist at the level of reference-fixing, while granting a substantial role to descriptive content in cognitive access, attribution, and transmission. The descriptive material does not define the referent — the referent was fixed at baptism, through a real R-grounding with a real individual — but it does play a specific role in how contemporary speakers anchor their use of the name. Proper names, as a category of linguistic item, are treated by competent speakers as tracking unique individuals who actually existed and were originally baptized; this categorial feature of names is what allows reference to persist after R is no longer available, and it is what distinguishes names from both descriptions and generic terms. The name makes possible a portable analogue of local rigidity in the absence of the individual.

The full development of this picture belongs to Chapter 5. For now, the compressed statement is: rigid designation can be approached as a social stabilization of the holding-fixed pattern first exhibited in local rigidity, and the dissertation's account explains both how the stabilization is achieved at baptism and how contemporary speakers continue to use the resulting names without direct access to the original coordination.

1.4. Four Converging Arguments

The thesis is supported by four converging lines of argument. Each makes visible the same presupposed mechanism from a different angle, and no single line is load-bearing—the thesis stands on their convergence.

The first line examines the formal logic of rigid designation. It reads Kripke's necessity-of-identity formula and shows that the formula's validity rests on the rigidity of variables under assignment, while the philosophical conclusion concerns ordinary names. Bridging the two requires the premise that names are rigid in the way variables are — a premise Kripke argues for separately, not one the formula delivers. The bridge is where a cognitive mechanism is needed and not supplied.

The second line traces the historical dialectic between Russell and Kripke. It shows that descriptions inherently fail to supply rigidity in modal contexts, and that Russell had actually identified the correct non-descriptive mechanism—demonstrative reference under acquaintance. Russell's error was restricting this mechanism to immediate sense data; freeing it from that Cartesian restriction provides the exact tool Kripke's semantics requires.

The third line applies this cognitive mechanism to classical modal puzzles. It argues that de re predication presupposes the ability to hold a particular fixed across modal variation. In perceptual-demonstrative cases this is supplied by x^o-mode coordination; in descriptive and mathematical cases the corresponding fixation may be A-grounded rather than R-grounded. It also clarifies the necessary a posteriori by showing that the necessity of identity depends on the fixed referent, while the a posteriority depends on the empirical route by which speakers discover that two designators converge on that referent.

The fourth line examines the actual metasemantic practice of naming. By analyzing an ordinary baptismal exchange and the subsequent causal chain, it shows that names attach to an already-active demonstrative coordination rather than producing rigidity *ex nihilo*.

The arguments are independent in the sense that rejecting any one leaves the others standing. They converge because they are symptoms of the same structure: something has to hold a particular fixed, and in each case, that something turns out to be a form of demonstrative coordination rather than formal apparatus or descriptive content.

1.5. Methodology and Scope

The dissertation is a work in the metasemantics of singular reference. It proceeds by close reading of Kripke's "Identity and Necessity" and *Naming and Necessity* in connection with the relevant passages in Russell, Wittgenstein, and Kaplan. The PRU framework (Cojocariu 2026) is used as an analytical tool; its full specification lives in a separate manuscript and is not defended here. What the dissertation uses of PRU is the distinction between object-mode (x^o) and concept-mode (x^c) of a word, the grounding operations described above, and the picture of baptism as the introduction of a new concept-word that inherits a live coordination. These pieces are introduced as they become needed rather than surveyed in advance.

The scope is singular reference. General-term rigidity — the question of whether and how "water" or "gold" are rigid — is a connected issue, and the object-mode / concept-mode distinction suggests a way of asking it, but the question is not pursued here. Fictional names, mathematical reference, and the broader modality spectrum are flagged in the conclusion as open questions rather than treated in the body. The diagnostic application of the A-/R-grounding distinction to Russell, Quine, and Lewis is suggested in this introduction and touched on in the positive account, but the full historical treatment is reserved for a separate paper. The point is not to offer a total theory of reference but to close one specific gap — the gap between Kripke's semantic account of rigidity and the cognitive operation it presupposes.

1.6. Roadmap

Chapter 2 develops the first line of support in depth. It reads Kripke's necessity-of-identity formula and its natural-language gloss, distinguishes three things the word "object" covers in that gloss — variable, encountered particular, named individual — and shows that the formula's validity holds at the first level while the philosophical thesis operates at the third. It connects the argument to Wittgenstein's treatment of identity in the *Tractatus* and shows that the formula is extensionally valid but intensionally silent. What the formula presupposes, when applied to names, is exactly what local rigidity describes.

Chapter 3 develops the second and third lines. It traces the Russell–Kripke dialectic, shows why descriptions cannot provide rigidity from within, and treats Kaplan's *dthat* as the clearest single piece of evidence that rigidity is prior to descriptions and not produced by them. The chapter closes by stating the explanatory gap that the remainder of the dissertation addresses.

Chapter 4 introduces the analytical core: the object-mode / concept-mode distinction and the formal development of A-grounding and R-grounding as operations of value-fixing within and beyond the linguistic system. It uses this machinery to clarify the cognitive-practical basis of de re readings and the necessary a posteriori, without replacing the standard scope-theoretic treatment of de re/de dicto ambiguity — the de re/de dicto distinction and the necessary a posteriori, treated as the level distinction applied to modal puzzles — and to treat an ordinary baptismal exchange as the observable form of what Kripke describes abstractly.

Chapter 5 gives the PRU account of rigid designation. It states formally what baptism does: the transition from a live demonstrative coordination to a coordination carried by a freshly introduced proper name. It distinguishes this transition from what happens after the baptismal community is gone, when later users operate through A-grounding with descriptive and testimonial material. It discusses the role of causal chains in transmitting this material, and closes with a brief coda placing the account in relation to the Kripke–Lewis disagreement about possible worlds.

Chapter 6 treats objections, including the objections that the A-/R-grounding distinction is a restatement rather than an advance, that the account collapses into sophisticated descriptivism, and that it relies on an unargued categorial feature of proper names. Chapter 7 concludes and marks open questions.

2. Identity, Objects, and Designators

Kripke's necessity-of-identity argument has two parts: a formal core, the modal logic of identity under fixed assignment, $\forall x \forall y(x = y \to \Box(x = y))$; and a metasemantic premise, that ordinary proper names function as rigid designators. The formal core delivers the modal consequence cleanly. The metasemantic premise carries the philosophical weight, and Kripke argues for it independently of the formula, on intuitive grounds — the counterfactual cases of *Naming and Necessity* and the Gödel/Schmidt thought experiment.[1]

This split is a standard reading in the literature, and Kripke himself frames the argument in these terms. The chapter takes this as its starting point and asks a finer question: what does the metasemantic premise commit us to, once we look closely at *how the formula is applied to natural-language cases?* The argument proceeds in three steps. The first specifies what the formal core does and does not establish. The second draws a distinction within the metasemantic premise itself: the word "object," used in Kripke's own gloss and throughout the secondary literature, runs together three different kinds of things, and the formula's validity holds for one while the philosophical conclusion is about a third. The third locates where a cognitive mechanism is needed if the metasemantic premise is to be more than a posit.

The chapter does not supply the mechanism. That is the work of Chapters 4 and 5. What it does is mark the place where the mechanism is needed, with enough precision that the proposal can later be evaluated against the right desideratum.

2.1. What the Formula Actually Does

The formal core of the argument is uncontroversial. Under any assignment function g in a Kripke model, if $g(x) = g(y)$, then in every world the model constructs, $g(x) = g(y)$. The assignment function does not vary across worlds in the standard semantics; once an assignment is fixed, the variables continue to pick out their assigned values across every world the operator $\Box$ quantifies over. The formula is valid by construction.

Two observations follow. They are unremarkable in themselves; together they specify what work the formula can and cannot do for the philosophical project.

The formula's validity rests on the rigidity of variables under assignment, which is a feature of the formal apparatus, not a feature of natural-language names. When Kripke applies the formula to names, he is applying a formal result whose validity holds for one kind of designator (variables under assignment) to another kind of designator (proper names). The application requires the assumption that names exhibit the same world-invariance variables have by construction. *This is the rigid-designation thesis itself, not a consequence of it.* Kripke himself distinguishes the rigidity of names — *de jure* rigidity, built into the kind of designator a name is — from *de facto* rigidity, which attaches to descriptions whose satisfiers happen to be necessarily fixed (such as "the least

[1] Kripke, *Naming and Necessity* (1980), pp. 100–105; "Identity and Necessity" (1971). The reading of the formula as a conditional whose force depends on the rigidity of its terms is standard; see LaPorte, "Rigid Designators," *Stanford Encyclopedia of Philosophy*, §1.1. The intuitive cases for the rigidity of names: *Naming and Necessity*, pp. 74–78 (Nixon), pp. 60–62 and 75–76 (Aristotle), pp. 83–87 (Gödel/Schmidt).

prime").[2] He is also explicit that the wide-scope behavior of descriptions in modal contexts is not what the rigid-designation thesis is asserting: the thesis is about the kind of designator names are, not about scope manipulation.[3]

The formula has no resources for the intensional dimension of the cases that motivate Kripke's project. Hesperus/Phosphorus, Benjamin Franklin and the inventor of bifocals, water and H_2O — these are cases where identity is empirically discovered and then recognized as necessary. The formula regiments the modal status once rigidity is granted; the epistemic route by which speakers reach the identity, and the cognitive capacities involved in using a name as rigid in the first place, lie outside what the formula can address. The formula is extensionally valid but intensionally silent.[4]

2.2. Three Levels of "Object"

The standard reading distinguishes the formal core from the metasemantic premise. The chapter adds a finer distinction within the metasemantic premise itself. When the formula is applied to ordinary identity claims, the word "object" — used in Kripke's own gloss and throughout the secondary literature — covers three distinct things, and the philosophical work happens at a different level than the formal validity does.

Domain elements are mathematical objects: points in a set, self-identical by set-theoretic construction, their identity invariant across any world the model constructs. The formula's validity operates at this level. Variables are assigned to domain elements; the assignment function is the device through which formal rigidity is achieved.

Encountered particulars are individuals as they figure in perception and re-identification: this dog here, that star there, the man one might have met. They are individuated through encounter; their identity across encounters is an epistemic achievement, with room for error. The early astronomer saw Hesperus in the evening sky and Phosphorus in the morning sky; the identity of the two had to be established, not given with the encounters. Encountered particulars are the bridge between the formal apparatus and the philosophical applications, because they have the concreteness that descriptions can apply to (so the cases Kripke discusses are intelligible at all) while also being the sort of thing that can bear a name.

Named individuals are bearers of proper names: Aristotle, Venus, Hesperus, Benjamin Franklin. They are individuated through baptism and tracked across absence through testimonial

[2]Kripke, *Naming and Necessity*, p. 21, fn. 21. The *de jure* / *de facto* distinction matters here because the formula's presupposition is *de jure* rigidity — rigidity by how the designator functions, not by metaphysical accident.

[3]Kripke, *Naming and Necessity*, p. 12, fn. 15. Kripke notes that taking a description outside the scope of a modal operator can produce something that behaves like rigidity, and is explicit that this is not what the rigid-designation thesis is asserting. The point is later developed by Kaplan's *dthat* operator, examined in Chapter 3.

[4]A constraint from the *Tractatus* bears on what kind of identity claim is at stake. Wittgenstein observes (5.5303) that "to say of two things that they are identical is nonsense, and to say of one thing that it is identical with itself is to say nothing at all." *Identity claims are claims about designators co-referring, not about objects themselves being one.* This is why the necessity-of-identity formula, applied to names, concerns the modal status of co-reference between designators rather than a metaphysical relation among objects — and it is co-reference between distinct designators that the empirical cases ("Hesperus is Phosphorus") turn on. The bearing of 5.5303 on Kripke's necessity-of-identity argument specifically is not much developed in the commentary, which has engaged Wittgenstein and Kripke mainly through rule-following.

networks. Their identity across different names is discoverable — the Hesperus/Phosphorus case is the paradigm. The philosophical thesis of rigid designation operates at this level: names hold their referents fixed across counterfactual variation, where descriptions could track different individuals in different worlds.

Kripke is of course aware that the formal result concerns variables under assignment, not names as such. The point, rather, is that the philosophical application requires a bridging premise: ordinary names must function with the relevant kind of referential stability if the formal result is to illuminate natural-language identity statements. My claim is therefore not that Kripke commits a simple slide, but that the cognitive realization of the bridge remains undertheorized.

What the passage does is move smoothly between these levels in English. The smoothness can obscure distinct explanatory levels. Each level handles part of the argument; no single level handles the whole. The metasemantic premise — that ordinary names exhibit the world-invariance variables have by construction — is what makes the move from the first level to the third coherent. The premise is supplied by the rigid-designation thesis, which Kripke argues for separately on intuitive grounds (the modal-intuition tests of *Naming and Necessity*, the counterfactual cases about Nixon and Aristotle).[5]

The three-level distinction is useful for the rest of the dissertation. The cognitive question — what enables speakers to use names as rigid designators — is a question about how the metasemantic premise gets satisfied at the level of named individuals. Local rigidity, developed in Chapters 4 and 5, is the proposed mechanism. It operates between encountered particulars and named individuals: it explains how a baptism can establish a name whose later use tracks the same referent across absence. The formal apparatus does not need a cognitive story; the philosophical application does.

2.3. The Cognitive-Mechanism Gap

What the standard reading leaves open is the cognitive question. Kripke argues that names are rigid; he does not argue that they are rigid because of any particular cognitive mechanism. The causal-historical sketch in *Naming and Necessity* — baptism fixes the reference, the name propagates through chains of communication — is offered explicitly as a "better picture" rather than as a theory. Kripke is unusually direct about this self-positioning: he is not, he says, "trying to give any sort of necessary and sufficient conditions for reference... in any rigorous sense"; he is "painting a general picture."[6] The picture describes the output of the relation between names and their referents; the cognitive mechanism that produces and sustains the output is not analyzed.

That a cognitive mechanism is needed at all, and that the available candidates may be too narrow, is something Kripke himself notices in passing. In "Identity and Necessity," he quotes Russell at length on demonstratives as the only true names — terms that "really just name an object without describing it" — and on acquaintance as the only ground secure enough for identity claims to be "exempt from all imaginable doubt." Kripke's gloss is that on Russell's view, only names of

[5]Kripke, *Naming and Necessity*, p. 48.

[6]Kripke, *Naming and Necessity*, pp. 93–94. The full passage: "I'm not, in fact, trying to give any sort of necessary and sufficient conditions for reference... in any rigorous sense" (p. 94); "I think it is a much better picture than the picture presented by a description" (p. 93).

immediate sense data, expressed by demonstratives like "this" and "that," qualify as genuine names.[7] The passage is not used by Kripke to endorse Russell's account; it is used to motivate the contrast with ordinary proper names, which are rigid in Kripke's sense but do not meet Russell's Cartesian standard. What the passage acknowledges, without developing, is that there is a candidate cognitive mechanism for rigidity — demonstrative grounding under acquaintance — and that Russell himself made it too narrow to cover the cases Kripke wants to cover. The mechanism Russell identified is the right kind of thing; the scope at which Russell allowed it is too restricted. Closing this gap — preserving Russell's mechanism while extending its scope — is the dialectical move that Chapters 4 and 5 will make explicit.

This gap is widely acknowledged in the secondary literature. As the *Stanford Encyclopedia's* discussion of rigid designation notes, Kripke's appeal to stipulation in the treatment of *de jure* rigidity is 'more like a promissory note than the satisfaction of an explanatory obligation.' [8] The promissory note is for an account of the mechanism by which names achieve, in actual linguistic and cognitive practice, what variables achieve by construction.

The literature has attempted to discharge this note in several ways: Evans (1982) defends a 'discriminating conception' requirement; Devitt (1981) offers a causal-descriptivist hybrid; and Recanati (2012) develops the cognitive account of singular thought into the Mental Files architecture. More recently, philosophers like Imogen Dickie (2015) have sought to ground singular reference directly in our cognitive capacities for perceptual attention and object-tracking. A full metasemantic account must integrate these insights. The present dissertation enters this space alongside, rather than simply against, these accounts, arguing that they can be unified by identifying a single primitive coordination that scales from a live demonstrative encounter to the social transmission of a name. The next chapter examines these foundations in detail.

[7] Kripke, "Identity and Necessity" (1971), p. 144. The Russell passage Kripke quotes is from *The Philosophy of Logical Atomism* (1918) and runs: "if we want to reserve the term 'name' for things which really just name an object without describing it, the only real proper names we can have are names of our own immediate sense data, objects of our own immediate acquaintance. The only such names which occur in language are demonstratives like 'this' and 'that.' "

[8] LaPorte, "Rigid Designators," §3.

3. From Russell to Kripke: The Argument for Rigid Designation

3.1. Overview

Chapter 2 closed at the point where the necessity-of-identity formula, applied to names, presupposed that names exhibit the same reference-invariance across worlds that variables have under assignment. The formula does not establish the presupposition; Kripke argues for it independently.

This chapter presents those independent arguments, lays out why descriptions, taken as descriptions, do not supply the kind of rigidity the arguments demand, and reconstructs a dialectical path on which Russell isolates a plausible non-descriptive mechanism of reference and then restricts its scope so sharply that it cannot underwrite ordinary public proper names. It closes by examining three leading cognitive implementations of direct reference — Evans, Recanati, and Devitt — and argues that each captures part of the explanatory burden while stopping short of an account of the capacity that would make rigid designation cognitively intelligible. The positive account is the work of Chapters 4 and 5.

3.2. Rigid Designation: Kripke's Independent Argument

Kripke's thesis is that proper names are rigid designators: a proper name designates the same individual in every possible world in which that individual exists. *Naming and Necessity* defends the thesis through three converging arguments, none of which depends on the necessity-of-identity formula. The formula regiments the modal consequence once rigidity is granted; the case for rigidity runs separately.

The modal argument turns on counterfactual intuition. Aristotle could have died in infancy, never taught Alexander, and never written the *Nicomachean Ethics*. In that counterfactual it is Aristotle who failed to do these things, not someone else. If "Aristotle" abbreviated "the teacher of Alexander who wrote the *Nicomachean Ethics*," then in a world where no one satisfied that description the name would refer to no one; if it referred to whoever did satisfy it, the counterfactual would be about someone else. Both readings clash with the intuition that the counterfactual is about the man Aristotle himself.

The epistemic argument turns on ignorance and error. A competent user of "Feynman" may know only that Feynman was a physicist, or may believe things about him that are false. On a descriptivist theory, such a user either fails to refer or refers to whoever fits her associated description. The sharpest version is the Gödel/Schmidt case: suppose the incompleteness proof we attribute to Gödel was in fact discovered by a man named Schmidt, and Gödel merely stole it. On a descriptivist theory built around "the prover of the incompleteness of arithmetic," "Gödel" would refer to Schmidt. The intuition is that "Gödel" still refers to Gödel. Reference runs through the name, not through the descriptions attached to it.

The semantic argument targets substitutivity in modal contexts. "Necessarily, Benjamin Franklin is Benjamin Franklin" is true; "Necessarily, the inventor of bifocals is the inventor of bifocals" is also true; but "Necessarily, Benjamin Franklin is the inventor of bifocals" is false — he might never have invented them. The name and the description share an actual-world referent but differ in modal profile. They cannot be synonyms.

Taken together, the three arguments support the distinction between de jure rigidity, characteristic of proper names and rigid by how the designator functions, and de facto rigidity, which can attach to descriptions whose satisfiers happen to be necessarily fixed across worlds, as with "the least prime." What they do not by themselves supply is an account of the cognitive operation by which speakers come to use a name in the rigid way the semantic thesis attributes to it.

3.3. Why Descriptions Cannot Supply Rigidity

Descriptions, qua descriptions, pick out satisfiers world by world. "The inventor of bifocals" picks out whoever, in a given world, invented bifocals; different worlds can therefore yield different referents. This world-relative behavior is constitutive of descriptive terms: they specify a condition, and reference goes wherever the condition is met. Adding further descriptive material may narrow the range of worlds in which the description is satisfied, but it does not by itself make the reference invariant across those worlds.

The formal expression of the difficulty is the scope problem. In $\Box\, F(\iota x\, Gx)$, the description $\iota x\, Gx$ can take scope inside or outside the modal operator. Narrow scope ("necessarily, F of whoever is G") requires that in every world the satisfier of G in that world is also F. Wide scope ("of whoever is actually G, necessarily F") requires that the actual-world satisfier of G is F in every world. The wide-scope reading behaves modally as if the description were rigid: the referent is fixed in the actual world and held constant. But the rigidity is positional, not constitutive — it is achieved by choice of scope, not by anything intrinsic to the description.

Something similar applies to Kaplan's dthat operator. dthat$(\iota x\, Fx)$ takes a description and yields a directly referring term whose reference is fixed in the actual world and then held constant across worlds. The result behaves rigidly, but the rigidity is introduced by the rigidifying device, not supplied by the descriptive content itself.

Names differ from rigidified descriptions in a further respect. A proper name does not contain descriptive structure that could generate a genuine narrow-scope reading of the ordinary descriptivist kind. "Aristotle" does not pick out whoever happens to satisfy some condition in each world; if it is rigid, it is rigid by the way it functions, not by scope placement or post hoc rigidification.

3.4. Russell's Mechanism and His Retreat

If descriptions cannot supply rigidity, the question becomes what non-descriptive cognitive operation can. In a passage from *The Philosophy of Logical Atomism* that Kripke quotes, Russell identifies the operation but immediately restricts its scope:

> "If we want to reserve the term 'name' for things which really just name an object without describing it, the only real proper names we can have are names of our own immediate sense data, objects of our own immediate acquaintance. The only such names which occur in language are demonstratives like 'this' and 'that.' "

Two claims are joined in the passage, and they must be separated.

The first claim is the isolation of a kind of non-descriptive reference. Russell singles out the demonstrative and characterizes it by what it does not do: it does not refer by way of a descriptive intermediary. In the vocabulary developed in Chapter 4, I will argue that this isolates the sort of operation later needed for local rigidity: direct, non-descriptive coordination with an experienced particular.

The second is the retreat. Having identified the mechanism, Russell restricts it to immediate sense data — the subset of cases in which the reference is exempt from Cartesian doubt. The restriction is epistemological, driven by a foundationalist demand for certainty. Under Russell's standard, even "this chair" is not a genuine name, because one could be hallucinating the chair.

The dialectical position of this dissertation begins here. Russell's demonstrative model identifies the kind of operation the rigid-designation literature seems to require: a way for a term to hold its referent fixed without reference being routed through identifying descriptive content. But by confining that operation to incorrigible cases of acquaintance, Russell leaves it unavailable for ordinary public proper names.

The restriction looks unnecessary once two properties are separated: directness and infallibility. For present purposes, a cognitive operation counts as direct when reference is not fixed by the mediation of identifying descriptive content; nothing in that notion by itself entails immunity to error. A direct operation may still misfire under sensory noise, contextual shift, or attentional lapse. Directness is thus a structural feature of the operation, whereas infallibility is an epistemic standard it may or may not satisfy. Once the two are prised apart, Russell's mechanism can be retained without his Cartesian restriction. Chapter 4 develops the cognitive details of how a direct-but-fallible operation could ground rigid reference to ordinary physical objects.

3.5. Baptism and the Causal Chain

Kripke's positive picture of how ordinary names attach to their referents runs through baptism and causal-historical transmission. At an initial act — the baptism — a name is attached to its bearer either by ostension ("let us call this child 'Feynman' ") or by a reference-fixing description ("let 'Jack the Ripper' denote whoever committed these murders"). From this origin the name propagates through a chain of communication: each speaker acquires the name from an earlier speaker, intending to use it with the same reference. The speaker at the end of the chain refers to the original bearer even if she knows nothing about him beyond the name.

Kripke is explicit that this is a "better picture" rather than a systematic theory:

> "I'm not, in fact, trying to give any sort of necessary and sufficient conditions for reference... in any rigorous sense."

The picture describes the external, social constraints on reference. It tells us what reference looks like from the outside; it does not describe the cognitive operations that realize the picture in actual speakers.

A reference-fixing description used in baptism — "let 'Jack the Ripper' denote the perpetrator of these crimes" — fixes the reference of the name but does not become its meaning. The name, once fixed, is rigid; the description is not. What is inherited down the chain is the designator and its rigidity, not the descriptive route by which the reference was originally established. This accounts well for speaker ignorance and error: a child can learn “Aristotle” from a parent and refer to Aristotle with no descriptive content beyond the name. What it leaves open is the cognitive level — how a brain actually performs the baptismal fixing and the chain-internal transmission.

What remains open, then, is not the semantic profile of rigid designation but its cognitive realization. Kripke’s picture explains how reference is socially stabilized and transmitted; it does not yet explain what capacity individual speakers exercise when they fix a referent in baptism, retain it across absence, and inherit it from others in a chain.

3.6. The Explanatory Gap

Joseph LaPorte characterizes Kripke’s appeal to stipulation as “more like a promissory note than the satisfaction of an explanatory obligation.” On the line pursued here, the promissory note concerns the cognitive operation that would underwrite rigid tracking, baptismal fixing, and chain-internal transmission. These need not be treated as three unrelated gaps; they may be understood as three roles that depend on a common capacity to single out a particular and continue to track it through change, across modal variation, and across handoffs between speakers. The semantic theory describes what the capacity delivers; it does not yet describe the capacity itself.

3.7. Existing Cognitive Implementations

There are several prominent attempts to fill Kripke’s cognitive gap, most notably Evans’s discriminating conception, Recanati’s mental files, and Dickie’s perceptual-attention framework. These accounts correctly recognize that reference requires a non-descriptive cognitive link to an object (what Recanati calls an ‘Epistemologically Rewarding relation’). The present account builds on this insight but shifts the focus: rather than treating reference-fixing primarily as a matter of internal mental representations or private files, it frames it as a publicly usable, dual-mode linguistic operation (x^o/x^c). The prior capacity to single out a particular in an encounter is what I will call *local rigidity*.

3.8. Summary and Transition

The chapter has argued for three connected claims. First, Kripke’s independent arguments support the rigidity of names, but they characterize a semantic property without yet explaining the cognitive operation by which speakers use names rigidly. Second, descriptions, taken as descriptions, do not supply constitutive rigidity, even though scope and rigidifying devices can produce rigidity-like behavior. Third, Russell isolates the kind of non-descriptive operation the theory appears to need, but restricts it too severely by tying directness to infallibility.

The proposal pursued in the next chapter is that a primitive tracking capacity — *local rigidity* — can be used to separate those two notions and to supply the missing cognitive layer. Chapter 4 introduces the analytical vocabulary in which that proposal is formulated: the object-mode/concept-mode distinction x^o/x^c, the grounding operator $\mathcal{R}$, and the apparatus needed to articulate the necessary a posteriori and de re modality in PRU terms.

4. The Functional-Mode Distinction

The present chapter introduces the analytical vocabulary in terms of which the rest of the dissertation closes the gap. The framework's basic primitives — the experiential pattern $\{X\}$, the linguistic pattern $\langle x\rangle$, and the coordination operator $\mathcal{R}$ — are set out in detail in the PRU framework document and are not duplicated here. The chapter develops what they support: a functional-mode distinction between x^o and x^c, a treatment of predication as coordination rather than function-application, the notion of local rigidity, and re-readings of two classical modal puzzles — the *de re*/*de dicto* distinction and the necessary *a posteriori*.

4.1. The x^o/x^c Functional-Mode Distinction

A word, on the account developed here, has two functional modes. As x^o, it coordinates with a particular instance of an experiential pattern $\{X\}$. As x^c, it operates within the linguistic pattern $\langle x\rangle$ — the network of utterances, frames, and inferential relations in which the word figures. The distinction is functional rather than lexical: the same word-form, "bird," can play either role, and which role it plays in a given utterance depends on what work the word is being asked to do.

Two examples isolate the modes. In "that bird is injured," "bird" functions as $bird^o$. It coordinates with one bird in the speaker's perceptual field. The utterance is about that bird; substitute another bird and the truth-value can change. In "birds fly," the word functions as $bird^c$. It operates within the inferential and generic relations the word maintains in $\langle bird\rangle$ — what is said about birds in the community, how the term combines with predicates, what counts as a counterexample. No particular bird is in view. The utterance is about how the word "bird" connects to "fly" within the linguistic pattern, with whatever generic force the form carries.

The two modes can be told apart by what speakers can and cannot do with them. A child who has acquired $bird^o$ can point at a bird and say "bird"; she can re-identify the same bird across short stretches of time. She cannot yet answer "what is a bird?" except by ostension. A child who has acquired $bird^c$ can answer in words: "a bird is an animal that flies." The difference is not in the word's reference but in the kind of work the word does — coordinating with a particular versus moving through a network of other words. Both are linguistic capacities. Neither reduces to the other.

This last point bears emphasis because it is where the framework most often gets misread. The object-mode x^o is *still language*. It is not a perception, not a piece of experience. Experience belongs to $E\{X\}$ — the activation, in a given encounter, of the perceptual-motor-affective constellation $\{X\}$. The word $bird^o$ does not *consist of* the experience of the bird; it coordinates with it. The two-side structure — $\langle x\rangle$ on the linguistic side, $\{X\}$ on the experiential side, $\mathcal{R}$ as the coordination between them — keeps x^o from collapsing into either pure linguistic relation or pure perceptual content. A speaker using a word in x^o-mode is doing something linguistic, but doing it in a way that requires coordination with what is encountered.

The closest structural analogue in the existing literature is Recanati's mental files framework. The present account builds on similar insights but differs in a crucial respect: it locates the distinction at the level of publicly usable word-modes and social-linguistic practices rather than treating it primarily as a classification of private mental vehicles. Furthermore, it applies this distinction symmetrically to common nouns and names.

4.2. $\mathcal{R}$ as Coordination

The relation $\mathcal{R}$ is the framework's coordination operator. It describes how the items of the two-side topology — words in $\langle x \rangle$ and patterns in $\{X\}$ — stand in relation to one another. The relation can be of several kinds, depending on what is being coordinated. A word in object-mode can coordinate with an experiential activation: $\mathcal{R}(bird^o, E\{BIRD\})$ describes the binding by which "bird" used demonstratively coordinates with a particular bird in view. A word in object-mode can coordinate with a word in concept-mode: $\mathcal{R}(bird^o, white^c)$ describes the predicative coordination by which "that bird is white" attaches the concept-mode predicate to the object-mode subject. A word in concept-mode can coordinate with another word in concept-mode: $\mathcal{R}(bird^c, fly^c)$ describes the generic coordination by which "birds fly" connects the two concept-mode items within the linguistic pattern.[9]

In each case the operator names the same kind of phenomenon — two items standing in coordination — and the argument types determine the specific kind. The claim of the framework is that what looks like several distinct things in standard semantics — reference, predication of a particular, generic predication, and baptism — are applications of one relation. The claim earns its keep only if the unification does work that separate treatments do not, and it does: because baptism $\mathcal{R}(h^o, N^c)$ and attribution $\mathcal{R}(\mathcal{D}, N^c)$ are the _same_ operator differing only in argument type, the account can say precisely where reference-fixing and description part company (the former crosses the experience–language boundary, the latter stays within $<x>$) —a distinction the reply to descriptivism in Chapter 7 turns on. A theory with four unrelated relations cannot locate the asymmetry in one place; this one can.

The word "coordination" carries the substantive load here. The standard alternative is Fregean function-application: a predicate is a function that takes an object as argument and yields a truth-value. Set-theoretic semantics produces a parallel architecture: a predicate is a set, predication is membership. Both pictures are asymmetric and arity-fixed: the predicate is the active item, the object the passive one, and the predication relation is the application of one to the other. The coordination picture differs at exactly this point. The two items in $\mathcal{R}(x, y)$ stand in a structural relation that the practice sustains, not in a function-argument pairing the speaker computes. The asymmetry between "subject" and "predicate" in surface grammar is real but not constitutive: in $\mathcal{R}(bird^c, fly^c)$ neither item is more "argument" than the other, and the same operator covers the case where one item is in object-mode and the other in concept-mode.

This is not a computational picture in an algorithmic or procedural sense. R is not an operation a speaker *performs* on linguistic items, in the way a function gets applied to an argument or a set-membership relation gets checked. Rather, $\mathcal{R}$ functions as a scale-free formal mapping

[9] $\mathcal{R}$ is specified in the framework document at greater technical length, including its developmental account in the learning of $\{X\}$ and its role in the integrated $\{X, x^o\}$ constellation produced by the coordination of experiential event and word; see PRU Meta-Language, version 6.

that tracks structural coordination across a cognitive continuum, spanning from sub-personal pattern-binding to socio-pragmatic embeddedness. To say that $\mathcal{R}(bird^o, white^c)$ holds is to say that the speaker's use of "white" for this bird is sustained by an objective convergence across multiple tiers of this continuum. It tracks an ongoing engagement with both the bird (perceptually, at the level of object-mode invariants) and the predicate (within the linguistic space $<white>$, which governs application conditions, inferential roles, and contrast cases). This coordination is constituted developmentally—through the agent's transition from basic tracking mechanisms to public practices—and is sustained by continued participation in the linguistic community and continued contact with the world. The register is closer to Wittgenstein than to cognitive engineering, provided $\mathcal{R}$ is understood as a scale-free formal mapping that tracks structural coordination across a cognitive continuum, rather than a computational procedure executed in real time. On this view, the high-level linguistic space $<x>$ is formed when language games are learned inside a practice. For example, $<cat>$ designates the totality of sentences using "cat" that make sense in a given language game, which $\mathcal{R}$ formally maps onto a corresponding practice with $\{CAT\}$, the lived cognitive attractor state. $\mathcal{R}$ thus functions as a formal descriptor of how language is lived, structurally anchoring public linguistic practices to sub-personal connectionist dynamics without collapsing into anti-mentalism.

4.3. Local Rigidity at x^o

Consider the utterance "that bird could have been larger." A speaker pointing at a bird in front of her makes the counterfactual claim. The bird in question is held fixed across the modal variation: the speaker is not imagining a different bird that is larger; she is imagining *that very bird* having been larger. The word "bird" in x^o mode coordinates with the same particular across the modal scope of "could have been." Whatever properties of the bird are varied — size, color, vigor, age — the bird stays the same, by virtue of the demonstrative coordination already in place.

The modal test for rigid designation is whether a designator picks out the same individual across the worlds the modal context quantifies over.[10] Within the modal scope of this utterance, the test passes: in every counterfactual scenario where "that bird could have been larger" is evaluated, "that bird" picks out the same bird that is in front of the speaker in the actual context. The variation occurs in what is predicated of the bird, not in which bird is being talked about. This is what rigid designation looks like at the level of demonstrative coordination.

Call this *local rigidity*: rigidity that obtains within a perceptual or conversational episode by virtue of the demonstrative coordination $\mathcal{R}(x^o, \cdot)$ being in place. Local rigidity is not yet Kripkean rigidity proper. Kripkean rigidity attaches to a name and holds across the entire range of counterfactual scenarios in which the bearer exists; local rigidity holds within the episode and lapses when the coordination does — when the speaker stops attending to the bird, or when the conversational context shifts. The two differ in scope and social scaffolding, while exhibiting a closely related holding-fixed structure. What is varied is the durability of the coordination, not the cognitive operation that constitutes it.

[10]Kripke, *Naming and Necessity*, p. 48.

This last point bears emphasis, because the structural claim of the chapter depends on it. Rigidity is not a stiff property of the designator. It is a preserved link between the speaker's use of the term and the particular that anchors the use. The preservation is constituted by the practice — perceptual attention, in the demonstrative case; social-linguistic stabilization, in the case of names. To call a term rigid is to say that this link is being preserved across whatever variation the modal context introduces. The variation is in the properties imagined, not in the link itself.

It is worth setting this picture against the natural alternative. A different way of conceiving rigidity treats the designator as something with rigidity built in — a kind of stiff attachment to its bearer that, once established, holds across modal variation by virtue of what the designator is. On that picture, what makes the modal evaluation come out right is the rigidity of the attachment itself, a feature the designator possesses. The picture is a natural one and is encouraged by some of Kripke's own formulations, but it misdescribes what is happening. Names do not have rigidity built in. Names rigidly designate because the linguistic community has a practice of using them rigidly — of correcting failures of reference, of holding the term to its bearer across descriptive revision, of teaching the term so that successive speakers continue to use it in the same way.[11] The preserved-link framing captures what this practice accomplishes; the built-in-property framing obscures it.

Names, on the account developed here, are devices for sustaining the preserved link across the conditions that make demonstrative re-anchoring impossible: the bearer is not perceptually present, the speaker is not the baptizer, the chain of transmission is many speakers and centuries long. The cognitive coordination at work in $\mathcal{R}(bird^o, \cdot)$ provides the model for the kind of coordination later sustained in absent-name use such as $\mathcal{R}(\sigma, Aristotle^c)$ at the level of a surrogate-based reading of Aristotle two millennia after his death.[12] What differs is the scaffolding that sustains the coordination: perceptual contact in the first case, testimonial networks and surrogate objects (manuscripts, portraits, citations) in the second. The cognitive operation is the same; the practice that preserves the link is different.

Names have been treated as indexicals/demonstratives before (Recanati, and others). Those are claims about names' semantic category; mine is a claim about the cognitive coordination any such semantics presupposes. So I don't compete with them — I sit underneath them.

One clarification forestalls a natural misreading. The semantic literature already distinguishes kinds of rigidity — obstinate from persistent rigidity (Salmon 1981), *de jure* from *de facto* (§ above), the wide-scope "rigidity" a description can be given by scope placement. Local rigidity is not another entry in that taxonomy. Those are classifications of how a designator behaves across worlds once it is in hand; local rigidity is a claim about the coordination that any such designator presupposes when it is anchored to a particular at all. It sits below the taxonomy, not beside it — the same relation to the rigidity literature that the account bears to Recanati's files at the level of

[11] Kripke's own metaphors are not in fact uniformly committed to a built-in-property reading. His description of the baptismal act and the causal chain (NN pp. 91–97) treats reference as something speakers do and sustain, which is closer to the preserved-link framing. The built-in-property reading is a tendency of the formal regimentation rather than of Kripke's positive picture; see Kripke's own characterization of his account as "a better picture" rather than a theory at NN p. 94.

[12] The notation σ for surrogate objects — manuscripts, portraits, citations — is from the PRU framework document.

singular thought.

The present account treats the demonstrative use of "bird" and the use of "Aristotle" as cases of closely related forms of coordination, differing in the practice that sustains the coordination rather than in the kind of cognitive operation involved.

4.4. *De Re* and *De Dicto*: Scope and the Holding-Fixed Operation

The *de re / de dicto* distinction is medieval in origin and gets one of its central modern formulations in Quine's puzzle about quantifying into modal contexts.[13] Consider the canonical example:

Necessarily, the number of planets is greater than seven.

The sentence is ambiguous. On one reading — *de dicto*, or narrow scope — "the number of planets" is evaluated at each world: in every possible world, the number of planets in that world must be greater than seven. The reading is false. On the other reading — *de re*, or wide scope — the description first picks out the number that is actually the number of planets, and the modal claim is then made about that number. Of that number, namely eight, it is necessarily true that it is greater than seven. The reading is true.

Two standard treatments handle this. Russell's scope-of-the-quantifier solution treats the ambiguity as one of logical form: the description can take scope inside or outside the modal operator.[14] Kripke's rigidity solution treats the relevant contrast as a feature of the difference between names and descriptions: names are rigid and behave like wide-scope descriptions in modal contexts; descriptions can be evaluated either way.[15] The present account does not replace these treatments. It adds a further question: what cognitive-linguistic operation allows a speaker to hold an object fixed for *de re* modal predication in the first place?

This is where the x^o/x^c distinction enters. It should not be read as a rival to scope theory. In the "number of planets" case, the fixing is not perceptual R-grounding. Numbers are not encountered particulars in the way birds, tables, or persons are. The object is secured by a formal or descriptive route, so the case is A-grounded rather than R-grounded. It is structurally analogous to holding something fixed, but it is not the paradigm case of x^o-coordination.

The clearer case is demonstrative:

That bird could have failed to fly.

Here the speaker is not saying that, in some possible world, whatever satisfies the description "bird in front of me" fails to fly. Nor is she making a general modal claim about birdhood. She is holding fixed this very bird and varying what is predicated of it. In the vocabulary of this dissertation, the use of "that bird" is anchored through $R(bird^o, E(\{BIRD_i\}))$, where $E(\{BIRD_i\})$ is the active encounter with that particular bird. This is the local form of *de re* predication: the particular

[13] Quine, "Reference and Modality," in *From a Logical Point of View* (1953); Kaplan, "Quantifying In," *Synthese* 19 (1968): 178–214.

[14] Russell, "On Denoting," *Mind* 14 (1905): 479–93, on scope generally; the modal application is developed in Smullyan, "Modality and Description," *Journal of Symbolic Logic* 13 (1948): 31–37.

[15] Kripke, *Naming and Necessity*, pp. 48–60 and the scope footnote at p. 12.

is fixed first; modal predication proceeds afterward.

Proper names can be understood as a socially stabilized extension of this structure under conditions of absence. Once a name has been successfully introduced and transmitted, its ordinary modal use resembles the *de re* side of the contrast: "Aristotle could have failed to teach Alexander" holds Aristotle fixed and varies what is predicated of him. But the current speaker does not perform fresh R-grounding with Aristotle. The name functions *de re* because a linguistic practice preserves the reference fixed by earlier acts of naming and transmission.

This also clarifies Quine's worry about essentialism.[16] The account does not dissolve the metaphysical issue. Holding an object fixed explains how *de re* modal predication is possible as a cognitive-linguistic practice; it does not decide which properties, if any, the object has essentially. It separates two questions: how a speaker can make a modal claim about a particular as that particular, and which modal claims about that particular are true.

The contribution of the x^o/x^c distinction is therefore modest but important. Scope theory tells us how the readings differ in logical form. The object-mode/concept-mode distinction explains how speakers are able to hold a particular fixed in the first place, and why demonstrative and name-based reference behave differently from purely descriptive or conceptual deployment.

4.5. The Necessary *A Posteriori* at the Level Distinction

Kripke's claim that some identities are necessary but *a posteriori* broke a connection that had been taken for granted in much of the analytic tradition: that necessity goes with analyticity and *a priori* knowability, while *a posteriori* truths are contingent. The classical examples are identities involving rigid designators — "Hesperus is Phosphorus," "water is H_2O," "this very table is made of wood." Each is necessary, on Kripke's account, because both flanking terms rigidly designate the same particular or natural kind. Each is *a posteriori* because the identity was established empirically.[17]

The phenomenon has been read by some as paradoxical. If "Hesperus is Phosphorus" is necessary, why was the identity not known *a priori*, like "2 + 2 = 4"? If it is *a posteriori*, how can it be necessary rather than contingent? The puzzlement is real on a picture where necessity is identified with analyticity and epistemic accessibility tracks metaphysical modality. On the picture developed in this chapter, the puzzlement has a clean source: it conflates two levels of language use that should be kept apart.

x^o supplies the fixed object whose identity grounds the necessity; x^c explains why recognition of that identity is empirical. When Hesperus is observed in the evening and Phosphorus in the morning, the same celestial body is the object of two demonstrative coordinations — once at dusk, once at dawn. At the level of x^o, there is only one Venus: $E\{VENUS\}$ activated in the evening is, as it happens, $E\{VENUS\}$ activated in the morning. The identity of the planet is constitutive

[16] Quine, "Reference and Modality" (1953); "Three Grades of Modal Involvement," *Proceedings of the XIth International Congress of Philosophy* 14 (1953): 65–81.

[17] Kripke, *Naming and Necessity*, pp. 100–105 on Hesperus/Phosphorus; pp. 116–18 on water/H_2O; the general thesis is at pp. 35–39. The pre-Kripkean connection between necessity, the *a priori*, and analyticity was defended in different forms by Carnap, Ayer, and others, and is the standard target of Kripke's first lecture.

at the demonstrative-coordination level: the same particular is being tracked in both cases. If the evening and morning observations do in fact concern the same celestial body, then the relevant identity lies at the level of the object being tracked, not at the level of the names or descriptions through which speakers conceptualize it. The empirical discovery is precisely the realization that two independently established modes of access actually converge on one and the same object.

The *a posteriority* lives at x^c. The concept-mode designators $\langle Hesperus \rangle$ and $\langle Phosphorus \rangle$ are distinct linguistic patterns. Each was established by an independent baptismal lifting: one anchored in the evening-sky coordination, the other in the morning-sky coordination. As concept-mode terms they have different usage histories, different inferential connections, different roles in their respective linguistic patterns. Establishing that they coordinate to the same particular requires empirical work — astronomical observation, theoretical inference, and acceptance of the identification by the community. The inference $\mathcal{R}(Hesperus^c, Phosphorus^c)$, that the two concept-mode patterns share a referent, is *a posteriori* in the standard sense.

The phenomenon is not a paradox on this reading. It is a structural fact about how language operates at two levels. The identity is necessary at the level where the same particular is being coordinated with; the discovery of the identity is *a posteriori* at the level where the two linguistic patterns are reconciled. The statement is one statement, but its modal and epistemic profiles are explained by different aspects of our practice: the fixed referent explains necessity; the historical/conceptual route by which speakers discover co-reference explains a posteriority. The traditional connection between necessity and *a priori* knowability assumed that there is only one level at which the statement operates. The x^o/x^c distinction blocks the assumption.

The same applies to "water is H_2O." At the level of x^o, the natural kind whose instances were demonstratively tracked by speakers using "water" is the same natural kind whose instances are chemically composed of H_2O molecules. There is one kind, and the demonstrative tracking by ordinary speakers and the chemical analysis by scientists coordinate to it. The identity is necessary at this level. At the level of x^c, the two terms — the ordinary-language $\langle water \rangle$ and the chemical-formula $\langle H_2O \rangle$ — are different linguistic patterns with different inferential roles. Establishing that they coordinate to the same kind required scientific investigation. The identity is *a posteriori* at this level.[18]

The necessary *a posteriori* has been handled in the literature through theories of trans-world identity, essentialism about substances and natural kinds, and two-dimensional semantics.[19] The present treatment does not contest any of these accounts at the semantic level. It locates the apparent paradox at the cognitive-linguistic level: necessity at x^o, *a posteriority* at x^c. The two levels were conflated in the tradition Kripke broke from, and they remain easy to conflate even after Kripke. The level distinction marks them apart and gives a structural account of why the necessary-*a posteriori* cases have the shape they have.

[18]The water-H_2O case raises additional questions about natural kinds, essence, and the role of scientific theorizing in fixing reference. These are discussed in Putnam, "The Meaning of 'Meaning'," in *Mind, Language and Reality* (1975), and have generated a substantial literature. The present section makes no commitment on the metaphysics of natural kinds; the level distinction handles the modal-epistemic shape of the case without taking a position on whether being H_2O is the essence of water in any deeper sense.

[19]Salmon, *Reference and Essence* (1981), and the survey in Soames, *Beyond Rigidity* (2002), chs. 7–9.

5. The PRU Account of Rigid Designation

The present chapter offers the constructive proposal. It does not contest Kripke's semantic results; it explains how speakers come to inhabit the models in which those results hold. The argument runs from x^o-mode tracking, through baptismal lifting, through testimonial transmission, to the description-attribution practices by which most of us actually engage with names whose bearers we have never encountered, and ends with the metasemantic positioning of PRU relative to standard Kripkean model theory.

5.1. The PRU Framework

Only what is needed for the argument will be introduced here. The full framework is developed in a distinct preprint (Cojocariu, 2026). Three notions matter.

A **pattern-constellation** $\{X\}$ is a unified attractor state in a cognitive system, sculpted by repeated encounters and integrating sensory, motor, affective, and interoceptive dimensions as a single learned configuration. We never encounter $\{X\}$ as such; we encounter particular instances. A pattern-recognition event $E(\{X\})$ is the system's falling into the $\{X\}$-basin in response to a partial cue. Recognition and reaction are aspects of one event, not a detection step followed by a separate response — this is the Pattern-Recognition Unity thesis from which the framework takes its name.

A **linguistic pattern-constellation** $\langle x\rangle$ is the analogous attractor in linguistic space — the standing pattern of uses, collocations, inferential connections, and discursive habits associated with a word. $\langle x\rangle$ is to language what $\{X\}$ is to experience.

Words can be employed in two different modes. As object-word, noted x^o, is learned as a motor pattern inside a constellation pattern $\{X\}$. At this stage the infant says "cat" when experiencing $\{CAT\}$ but cannot explain what a cat is. Later on, immersed in the language game of adults, the kid learns uses of "cat" in sentences when there is no cat present. This use is called concept-word, noted x^c. When asked what a cat is, the kid can say "A cat is a furry animal", using cat^c.

The **grounding operator** $\mathcal{R}$ coordinates these two kinds of constellation. At the surface level we write $\mathcal{R}(x^o, x^c)$ — relating the word in its object-mode use to its concept-mode entry. At the deeper architectural level the same operation is $\mathcal{R}(\{X\}, \langle x\rangle)$ — coupling the experiential basin to the linguistic pattern. The two notations refer to the same relation under different resolutions.

That is the apparatus. The remainder of this chapter is what it does.

5.2. Local Rigidity: x^o as the Language Primitive

Russell's acquaintance was meant to identify a special form of reference: a contact with an object that secures identification independent of any descriptive material. He restricted it to immediate sense-data because only there did he find the Cartesian certainty he took such reference to require. Once that restriction is dropped — once we allow that reference can be stable without being incorrigible — Russell's mechanism generalizes. What he saw in the sense-datum case is what happens whenever a speaker tracks a particular through encounter: a word in x^o-mode coordinates with an experiential instance, and the coordination is rigid for the duration of the tracking. "That bird could have been larger" requires me to hold the bird fixed across the modal evaluation; the holding is accomplished by my present perceptual contact, not by any description I could supply.

x^o is the PRU implementation of acquaintance with two specific modifications. PRU's grounding is **fallible**: $E(\{X\})$ can misfire, the basin $\{X\}$ is reshaped by later encounters, and I can be wrong about properties of what I am tracking. PRU's grounding is also **dynamic**: the attractor state is sculpted continuously, not given once and for all. These features make x^o less like Russell's punctual sense-datum and more like the standing capacity of a speaker to keep a particular fixed through engagement with it. Kripke's own account already permits the speaker to be wrong about the bearer of a name; the rigidity does not depend on infallibility. PRU pushes this point further into the cognitive substrate: tracking does not require certainty, only stability.

The cognitive primitive is therefore the x^o-coordination—the word in object-mode locked onto an experienced instance. At the primitive level, this coordination exhibits the clearest case of non-descriptive holding-fixed; it provides the basic model from which the later account proceeds. It is not a property added to reference; it is what reference is at its primitive level.

5.3. The Problem of Absence

The primitive form of reference has a limitation built into it. When the bearer leaves the perceptual field, the x^o-coordination dissolves. "That bird" stops functioning when the bird flies away. What had been a rigid grip becomes just a token.

The natural descriptivist response is to substitute description for tracking: in the absence of the bird, refer to it as "the bird I saw earlier in the willow." But Chapter 3 has already shown that descriptions cannot do the work of names. They pick out world-by-world satisfiers; they fail in modal contexts; they are constitutively x^c and have no x^o anchoring of their own to draw on. A speaker who has only descriptions to work with has, in the relevant sense, lost the particular.

This is the problem names solve. A name is a device for extending, under social conditions, the holding-fixed function first visible in x^o-coordination into the space of absence. The rigidity is real where the speaker is present; what is needed is a way to carry it forward when the speaker is not. Names are best understood as portable demonstratives: linguistic artifacts engineered to maintain rigid tracking when perceptual contact is no longer available.

5.4. Baptism as Lifting

A baptismal act exploits an existing x^o-coordination to inaugurate a new linguistic pattern. At the moment of baptism, some speaker is in perceptual contact with the individual to be named. $\mathcal{R}(human^o, E(\{HUMAN\}))$ is in place, I experience a particular human, this human; the object-mode word "human," or "child," or "this one," is rigidly coordinated with the particular instance. The baptism does two things simultaneously: it introduces a new concept-word N^c (the name), and it forms a coordination between the already-rigid h^o and this newly introduced N^c:

$$\mathcal{R}(h^o, N^c)$$

for example:

$$\mathcal{R}(h^o, {N^c}_{aristotle})$$

This effectively states, 'Let this human be named Aristotle.' But because proper names are introduced as standing linguistic devices rather than as passing demonstratives, what this accomplishes is practical rather than metaphysical. A new linguistic pattern $\langle aristotle \rangle$ enters the community's vocabulary. Its standing function — fixed by the baptismal context and maintained by subsequent use — is to track the same individual the demonstrative grip was holding. The rigidity of h^o in encounter is **lifted** into N^c. Not transferred and not duplicated: lifted in the sense that what was confined to the duration of perceptual contact becomes available for use beyond it.

This is why it is not baptism itself that creates rigidity. The rigidity was already there in $\mathcal{R}(h^o, \{HUMAN\})$. What baptism does is change its locus: from a fleeting demonstrative coordination to a durable, socially maintained linguistic pattern. The name inherits its modal profile from the demonstrative act at its origin. Specifically, it inherits scope-ambiguity-freedom: where a description has descriptive structure that interacts with modal operators in two ways, N^c has no such structure to interact with at all. It is rigid because the act that founded it had nothing variable to begin with.

Kripke's "initial baptism" is the same event described from the outside. What his account leaves implicit — that some speaker had to be tracking the individual at the moment the name was introduced — is what PRU makes explicit. The picture is not new; the cognitive precondition is.

5.5. Causal Chains as Testimonial Networks

The question is how rigid tracking propagates through a community when only the first speaker(s) had x^o-access. What is the mechanic of the causal chains that Kripke mentions?

Three transmission channels operate. **Linguistic patterns** $\langle N \rangle$ carry the bulk of the work: a speaker who has heard "Aristotle" used in the standard ways forms her own $\langle aristotle \rangle$ — a stable pattern of uses, collocations, and inferential connections that mirrors the community's. **Testimonial patterns** $\tau(N)$ are the statements, texts, narratives, and corrections that flow through the network: an academic conference, a textbook, a teacher's correction of a student's misuse. These shape and stabilize $\langle N \rangle$ for the speaker who internalizes them. **Surrogate constellations** $\sigma(N)$ are the residual experiential anchors: manuscripts, busts, recordings, photographs — objects that causally trace to the original bearer and provide a partial $E(\{X\})$ for speakers who encounter them. Surrogates are not necessary for reference. A blind historian uses "Aristotle" correctly without ever encountering an image. But for most speakers, surrogates supply the residual phenomenological connection that pure testimony cannot.

What is transmitted through these channels is N^c — not the bearer, and not any particular description. The name-concept retains its standing function because the community treats it as a tracking device rather than a classifier. This is the practice in which rule-following becomes intelligible (Wittgenstein, *Philosophical Investigations*, §§198–202): the rigidity is constituted by the community's collective treatment of $\langle N \rangle$, sustained by training, correction, and institutional embedding.

This is what Kripke's causal-historical chain comes to in cognitive terms. The chain is not a metaphysical wire between baptism and current use; it is a pattern of practices that keeps N^c doing its tracking work across generations.

5.6. Attribution after Transmission

A downstream speaker who acquires N^c through the causal chain—but who lacks direct perceptual access to the bearer—is in an asymmetric position. The name in her vocabulary inherits its rigidity from a baptismal coordination she never performed. Her cognitive engagement with the bearer must rely on a different mechanism. What is she actually doing when she uses the name?

The classical descriptivist answer is that she uses the name as shorthand for the descriptions she associates with it. As Kripke demonstrated, this fails because it leaves reference hostage to the accuracy of those descriptions. The present framework offers a structurally different answer: descriptions do not constitute reference; rather, they are attributed to a name-concept that already carries a fixed reference.

Let $\mathcal{D}$ denote a description, formally an ι-structured compound within $\langle x \rangle$-space. For Aristotle, a contemporary scholar might form several such descriptions:

$$\mathcal{D}_1 := \iota x\,(x \text{ taught Alexander the Great})$$

$$\mathcal{D}_2 := \iota x\,(x \text{ wrote the } \textit{Nicomachean Ethics})$$

$$\mathcal{D}_3 := \iota x\,(x \text{ studied under Plato})$$

Each of these is a $\langle x \rangle$-side object: a linguistic pattern with a uniqueness presupposition, evaluable at any world to yield a satisfier (or to fail). The scholar's cognitive engagement with Aristotle consists in coordinating these descriptions with the inherited name-concept:

$$\mathcal{R}(\mathcal{D}_1, N^c_{Aristotle}) \wedge \mathcal{R}(\mathcal{D}_2, N^c_{Aristotle}) \wedge \mathcal{R}(\mathcal{D}_3, N^c_{Aristotle}) \wedge \dots$$

Each conjunct is an **attribution**: a coordination running from a description to a name-concept. Several features of this structure matter.

The operator is the same $\mathcal{R}$ used throughout. There is no separate "attribution operator." What distinguishes attribution from baptismal grounding is the type of its arguments. Baptismal grounding is $\mathcal{R}(h^o, N^c)$ — an x^o-side argument coordinated with a $\langle x \rangle$-side argument, crossing the experience-language boundary. Attribution is $\mathcal{R}(\mathcal{D}_i, N^c)$ — both arguments live in $\langle x \rangle$-space, and the coordination stays within language. The operator's form is constant; what changes is the work it does, depending on what it is applied to.

The conjunction is the right form. Each attribution is independently established and independently revisable. Discovering that bifocals were not in fact invented by Franklin requires me to drop $\mathcal{R}(\mathcal{D}_{\text{bifocals}}, N^c_{BF})$. The other attributions — that Franklin signed the Declaration, that he was Postmaster General — remain intact. My reference to Franklin is not disturbed by the revision because reference was never carried by any attribution. Reference is carried by N^c, anchored at baptism by $\mathcal{R}(h^o, N^c_{BF})$, propagated through the chain. The attributions are how I cognitively inhabit the name, not how the name secures its bearer.

For convenience we may write the shorthand $\mathcal{R}(\mathcal{D}, N^c)$ when the internal structure does not need to be explicit. This abbreviation must be read with care: it stands for a conjunction whose individual conjuncts can be dropped, replaced, or revised without invalidating the others. The shorthand $\mathcal{D}$ is a structured collection, not a monolithic description.

Two consequences follow.

The asymmetry between rigid names and non-rigid descriptions becomes formally visible. $N^c_{Aristotle}$ is rigid in virtue of the baptismal $\mathcal{R}(h^o, N^c_{Aristotle})$ it inherits. Any $\mathcal{D}_i$ remains non-rigid: it picks out world-by-world satisfiers, and in worlds where Aristotle did not study under Plato, $\mathcal{D}_3$ either fails or picks out someone else. The attributive $\mathcal{R}(\mathcal{D}_3, N^c_{Aristotle})$ is itself a contingent claim about the actual world. Each attribution can be false. None of these contingencies touches the rigidity of $N^c_{Aristotle}$.

The Hesperus/Phosphorus case acquires a transparent structure. Before the astronomical discovery, two name-concepts had been independently propagated, each with its own attributions:

$$\mathcal{R}(\mathcal{D}_{\text{morning star}}, N^c_{\text{Phosphorus}})$$

$$\mathcal{R}(\mathcal{D}_{\text{evening star}}, N^c_{\text{Hesperus}})$$

The discovery is the recognition that N^c_{Hesperus} and $N^c_{\text{Phosphorus}}$ are inherited from causal chains that converge on the same baptismal act — that the two name-concepts are, in the relevant sense, one. The fact that the same star is baptized with different names becomes:

$$\mathcal{R}(star^o, N^c_{\text{Phosphorus}}) \wedge \mathcal{R}(star^o, N^c_{\text{Hesperus}})$$

The post-discovery configuration recognizes a single anchor under two inherited names:

$$\mathcal{R}(star^o, N^c_{\text{Hesperus}}) \wedge \mathcal{R}(star^o, N^c_{\text{Phosphorus}}), \quad N^c_{\text{Hesperus}} = N^c_{\text{Phosphorus}}$$

Two attributions, one name-concept. But writing $N^c_{\text{Hesperus}} = N^c_{\text{Phosphorus}}$ is just a logical consequence, and does not make sense in natural language use; both names will be the same and the logical sollution is to replace them with another name, in this case “Venus”: $N^c_{\text{Hesperus}} = N^c_{\text{Phosphorus}} = N^c_{\text{Venus}}$\$ The necessary a posteriori falls out structurally. Necessity comes from the $\mathcal{R}(h^o, N^c)$ stratum; a-posteriority comes from the $\mathcal{R}(\mathcal{D}, N^c)$ stratum; both strata are present in any contentful use of the name.

This is also what allows the cognitive engagement of a contemporary scholar with a long-dead figure to be substantive without being descriptivist. The scholar engages with Aristotle through her attributions. The attributions can be revised, refined, defended, or dropped. Through all such revisions, she continues to refer to the same individual — because $N^c_{\text{Aristotle}}$ continues to do its rigid work, independent of what she attributes to it.

5.7. PRU as Metasemantics

The standard Kripke model $\langle W, R, D, \mathcal{I} \rangle$ assigns to each name a denotation that is constant across worlds: $\mathcal{I}(N) = d$, where $d \in D$. The rigidity of names is built into the interpretation function. The model does not say how a real speaker comes to use a name in conformity with such an interpretation.

PRU does not propose a different model. It does not contest the formal semantics of rigid designation. What it offers is an account of how speakers come to inhabit Kripkean interpretations: how the rigidity that the model stipulates as a formal constraint is achieved in cognitive practice. The developmental trajectory has four stages:

The x^o-coordination is the cognitive primitive, present already in pre-linguistic tracking and in the demonstrative uses of common nouns. The baptismal lifting $\mathcal{R}(h^o, N^c)$ is the act by which a name-concept inherits the rigidity of its founding demonstrative grip. The testimonial transmission carries N^c across speakers and generations without requiring further x^o-access. The attributive coordination $\mathcal{R}(\mathcal{D}, N^c)$ is the cognitive mode in which most speakers, most of the time, engage with names whose bearers are absent or have always been absent.

The Kripkean model captures the output. PRU describes the process by which an agent comes to participate in that output. The two are not in competition. They operate at different levels of abstraction. The model stipulates $\mathcal{I}(N) = d$; PRU explains how a speaker is positioned to use N such that the stipulation correctly describes her semantic practice.

This is the sense in which PRU is metasemantics, not rival semantics. PRU supplies what Kripke's apparatus presupposes — the cognitive precondition Russell located and retreated from — without contesting the formal semantics.

The argument is now in place. The objections that this account is bound to attract are taken up in Chapter 6.

6. Objections and Replies

The account is now in place: rigid designation is the social stabilization of local rigidity, baptism lifts an x^o-coordination into a portable N^c, and contemporary speakers engage absent bearers through revisable attributions $\mathcal{R}(\mathcal{D}, N^c)$ that ride on a reference the attributions do not constitute. An account of this shape attracts objections of three kinds. The first says it has not added anything: that the A-/R-grounding distinction relabels a contrast already drawn. The second says it has added the wrong thing: that for all the anti-descriptivist language, the treatment of absent reference is descriptivism with extra notation. The third says it has helped itself to what it needed to prove: that the persistence of reference across absence rests on an unargued categorial feature of names. A fourth, more local, presses on the claim that x^o is a mode of *language* rather than a relabelled perceptual relation — the claim on which the whole "scaling up" of demonstrative rigidity depends. I take them in that order. None is a strawman; the first three are the objections I would raise against this account if someone else had written it, and the fourth is the one a careful reader reaches by the end of Chapter 5.

6.1. The Restatement Objection

The objection. The distinction between A-grounding and R-grounding, the critic says, is the old distinction between reference-by-description and reference-by-acquaintance wearing new clothes. Russell drew it. Evans redrew it as the contrast between descriptive and demonstrative thought. Recanati's typology of files is built on it. Renaming the descriptive side "A-grounding" and the demonstrative side "R-grounding," and the demonstrative-rigidity phenomenon "local rigidity," does not advance the debate; it supplies vocabulary. The thesis, on this reading, is a notational variant of a position already on the table, and its appearance of novelty comes entirely from the new symbols.

Reply. The objection would land if the contribution were the distinction. It is not. The acquaintance/description contrast is indeed old, and the dissertation says so: Russell drew it and then misplaced its boundary. What the account adds is not a new name for the contrast but a claim about its *direction of explanation* — and that claim is substantive, falsifiable, and absent from the views the objection cites.

The claim is that rigidity flows *upward* from demonstrative coordination to names, rather than belonging to names in the first place and being *approximated* by demonstratives. On the acquaintance tradition, the demonstrative case and the name case are two species of singular reference, sitting side by side; acquaintance is one route to a singular thought, description another. On the present account they are not coordinate species. Local rigidity is the primitive, and Kripkean rigidity is what local rigidity becomes when a social practice carries it past the encounter. This is a claim of derivation, not of classification, and it has consequences the classificatory view does not have.

Two of those consequences are testable against the rival's own commitments. First, on the derivation claim, common nouns used demonstratively ("that bird") must already exhibit rigidity, because the rigidity of names is built from exactly that material. The acquaintance tradition, focused on singular thought about individuals, has no reason to expect rigidity at the level of a kind-term used to pick out a passing instance; the present account requires it (Ch. 5, "Local Rigidity at xo" and "De Re and De Dicto"). Second, on the derivation claim, the failure mode of a name should be a failure of the *practice* that sustains the lifted coordination — a broken chain, a corrupted transmission — and not a failure of the speaker's descriptive knowledge. That is precisely the Feynman and Gödel data Kripke used against descriptivism, now predicted rather than merely accommodated. A relabelling predicts nothing. This account predicts both, and would be refuted if demonstrative kind-reference turned out to admit scope ambiguity, or if reference tracked descriptive competence rather than chain-membership.

So the distinction is old and the dissertation neither claims nor needs otherwise. What is new is the order of priority among its terms, and the order of priority does work the side-by-side picture cannot do.

6.2. The Sophisticated-Descriptivism Objection

The objection. This is the sharper charge. The account concedes that contemporary speakers have no R-grounding with Aristotle, that they operate through A-grounding, and that what they cognitively possess is a bundle of descriptions $\mathcal{D}_\infty$, $\mathcal{D}_\in$, $\mathcal{D}_\ni$ coordinated with the name (Ch. 5, "Attribution after Transmission"). But this is what descriptivism always said: that competent use of a name consists in associating descriptions with it. Dressing the bundle as a conjunction of revisable attributions $\mathcal{R}(\mathcal{D}_\rangle, N^c)$, and insisting that the descriptions are "attributed to" rather than "constitutive of" the referent, is a distinction without a difference. If you ask what, in the contemporary speaker's head, fixes which individual she is talking about, the only available answer on this account is: the descriptions. So reference *is* carried by the descriptions, and the anti-descriptivist protestation is empty.

Reply. The objection mislocates where reference-fixing happens, and this mislocation is undoubtedly what the account is built to expose. The question 'What, in the later speaker's mind, fixes the referent?' falsely presupposes that reference is fixed by the current user. On the present account, it is not. Reference was fixed once, at baptism, by a demonstrative coordination that downstream speakers neither performed nor can perform. What is in her head is not a reference-fixing device but an *inheritance*: the name-concept N^c, acquired through the chain, already carrying the reference that the baptismal $\mathcal{R}(h^o, N^c)$ established. The descriptions she attaches to N^c are how she cognitively inhabits an inheritance she did not author. They are not a mechanism for selecting a referent; they are what she has come to believe about a referent already selected for her.

This is not a verbal difference, because the two pictures come apart on the data. Take the case the objection cannot absorb: the speaker whose every description is false. Suppose a student associates with "Aristotle" only "the author of the *Nicomachean Ethics*" and "Plato's most famous pupil," and suppose — counterfactually — that scholarship overturns both, showing the *Ethics* was compiled by students and that the famous pupil was someone else. On descriptivism, the student's "Aristotle" now refers to whoever satisfies the surviving descriptions, or to no one. On the present account, the student still refers to Aristotle, because reference rode on N^c the whole time and N^c traces through the chain to the baptismal grounding, not through the student's beliefs. The descriptions were attributions; attributions can be wholly false without disturbing reference, exactly as a person's reputation can be entirely mistaken without making the reputation about someone else. Descriptivism cannot say this. The account predicts it. That is the difference the objection says does not exist.

The deeper point is structural. Descriptivism makes the descriptions do two jobs at once — fix the referent *and* express the speaker's understanding of it — and the anti-descriptivist arguments (Kripke's Feynman, Gödel) are arguments that the descriptions cannot do the first job. The present account agrees they cannot, and so it removes the first job from the descriptions entirely and assigns it to the baptismal grounding transmitted as N^c. The descriptions keep only the second job. A view that gives descriptions one of descriptivism's two jobs is not descriptivism, any more than a view on which a portrait resembles its sitter without determining who the sitter is collapses into the view that the sitter just is whoever the portrait resembles.

One residue should be granted rather than papered over. Reference rides on N^c through the chain — but a later speaker's grip on *which* chain her use of "Aristotle" belongs to is itself secured partly through testimonial and descriptive material ("the Aristotle my teacher meant," "the philosopher, not the shipping magnate"). The descriptivist can relocate the worry here: not at reference-fixing, but at chain-individuation. This is a real cost, but a smaller one. Disambiguating which chain one is on is an epistemic task that draws on description; it does not make description constitutive of the reference the chain carries, any more than needing a catalogue number to find a painting makes the number constitutive of which painting it is. The worry moves from the center of the account to its edge, and changes from "reference is descriptive" to "chain-identification is sometimes descriptively mediated," which the account can grant without giving up anti-descriptivism about reference itself.

6.3. The Unargued-Categorial-Feature Objection

The objection. The account's treatment of absence (Ch. 5, "Causal Chains as Testimonial Networks" and "Attribution after Transmission") rests on a claim stated in the introduction and never defended: that proper names, *as a category of linguistic item*, are treated by competent speakers as tracking unique individuals who actually existed and were originally baptized. This categorial feature is what carries reference past the point where R-grounding lapses. But the feature is exactly what needed explaining. To say that names persist in reference because they belong to a category whose members persist in reference is to name the explanandum and call it an explanans. The wheel turns nothing.

Reply. The charge is fair as stated, and meeting it requires saying what the categorial feature *is* and why positing it is not circular. The feature is not a brute semantic stipulation that names refer rigidly — that would indeed be the explanandum relabelled. It is a fact about a *practice*: the community treats name-tokens as governed by a norm of co-reference with their baptismal origin, and enforces that norm through correction. When a child uses "Aristotle" of the wrong figure, she is corrected, and the correction does not cite descriptive failure ("Aristotle wrote no such thing") but identity failure ("that's not who 'Aristotle' picks out"). The categorial feature is the standing of this norm in the practice, and it is no more circular to appeal to it than it is circular to explain the rigidity of a measurement standard by appeal to the institution that maintains it.

What keeps the appeal non-circular is that the norm has an *independent characterization* and an *independent origin*. Its characterization is behavioral and observable: it is the pattern of correction, deference, and chain-preserving transmission that Kripke described from the outside and that Chapter 5 redescribes in cognitive terms. Its origin is the baptismal R-grounding: the norm of co-reference-with-origin can attach to a name precisely because there *was* an origin — a real coordination with a real individual — for the norm to point back to. A category of terms with no such origins (genuinely empty names aside, flagged as an open case in the Conclusion) could not sustain the norm, which is why descriptions, which have no baptismal origin, do not behave as names do. So the categorial feature is not posited to explain rigidity and then used to explain rigidity. It is identified independently — as a correction-practice with a baptismal anchor — and *then* shown to deliver persistence across absence. The explanation runs from practice-plus-origin to persistence, not from persistence to itself.

I do not claim this dissolves every form of the worry. An objector may press: why does *this* practice exist, why do human communities maintain origin-tracking norms for names but not for descriptions? That is a genuine question, and it is anthropological and developmental as much as philosophical; the account locates the answer in the usefulness of carrying a fixed grip past the encounter (Ch. 5, "The Problem of Absence") but does not pretend to a full genealogy. What the reply establishes is narrower and sufficient for the thesis: the categorial feature is a describable practice with a describable anchor, not a synonym for the conclusion.

6.4. The "x^o Is Just Perception" Objection

The objection. The whole construction depends on x^o being a mode of *language* — a word-use — rather than a relabelled perceptual or causal relation between a speaker and an object. If x^o is really just perception (or perceptual attention, or a causal-informational link), then the account is a causal theory of reference in disguise, and the "coordination" $\mathcal{R}(bird^o, E(\{BIRD\}))$ is just the old causal relation with a new symbol. The claim that x^o "is still language, not perception" (§5.1) is asserted there but not argued, and the objection is that it cannot be argued, because there is nothing for x^o to be *between* perception and inferential role.

Reply. x^o is distinguished from perception by a double dissociation, and the dissociation is what shows it is a third thing rather than either pole. On one side, the experience can occur without the word: $E(\{X\})$ — the perceptual-motor-affective activation — happens in pre-linguistic infants and in animals with no word at all. Perception does not require x^o. On the other side, the word can occur, in object-mode, and *misfire* on its object: a speaker says "that bird" while pointing, and coordinates with the wrong thing in the visual field, or with a leaf she has mistaken for a bird. The word's object-mode use can be wrong about what it coordinates with, which a perceptual state cannot be in the same way — the perception is whatever it is; the *word applied to it* is what bears correctness. That a use can be incorrect is the mark of the normative, and the normative is the mark of language. So x^o is not perception (perception runs without it) and not free of the world (it can misfire against the world), and the space between those two facts is exactly the space x^o occupies: a public, correctable word-use that requires coordination with what is encountered but is not identical to the encounter.

This also answers the causal-theory charge. A causal-informational link between word and object is not the sort of thing that can be *mistaken* while remaining the same link; it either obtains or does not. x^o can be exercised incorrectly — wrong instance, wrong sortal — and be recognized as incorrect by the speaker and the community, and corrected. Correctability is not a feature of causal relations; it is a feature of moves in a practice. The coordination R is therefore not the causal relation relabelled, because it carries a normative status the causal relation lacks. What x^o shares with the causal theory is only the insistence that reference reaches outside the linguistic network; what it adds, and what makes it a *linguistic* mode rather than a causal hook, is that the reaching-out is something one can do correctly or incorrectly.

6.5. Two Smaller Objections

Demonstrative misfire and the rigidity claim. If x^o can misfire (Ch. 5, "Local Rigidity: xo as the Language Primitive"), how can local rigidity be *rigid*? A designator that might be coordinating with the wrong object hardly seems to "hold its referent fixed." The reply is that rigidity and infallibility are the two properties the dissertation has been at pains to separate since §4.4. Local rigidity is rigid in the modal sense — *given* what the speaker is coordinating with, every counterfactual she builds is about that thing — without being infallible in the epistemic sense about *which* thing that is. The misfire is a mistake about the target; the rigidity is a feature of how the target, once fixed, behaves under modal variation. These are orthogonal, and conflating them is the very error that drove Russell to immediate sense-data.

The status of surrogate objects. Chapter 5 grants that a blind historian refers to Aristotle with no surrogate σ, which seems to make σ idle. The reply is that σ is not part of the reference-fixing mechanism and was never claimed to be; it is one of three transmission channels, and the weakest. Testimony τ and the linguistic pattern $\langle$N$\rangle$ carry reference; surrogates only enrich the speaker's phenomenological relation to the bearer. The blind historian shows that σ is dispensable, not that it is doing covert work. An account on which surrogates were load-bearing would be embarrassed by the blind historian; this account predicts her, because it locates reference in the chain and the norm, not in any residual perceptual contact.

6.6. What the Objections Leave Standing

The four principal objections share a structure: each assumes that the account must locate reference-fixing *in the contemporary individual speaker* — in her distinction-drawing, her descriptions, her perceptual link, or a brute semantic feature she deploys. The thesis's recurring reply is that reference-fixing happened *once*, at a baptismal R-grounding, and that everything the contemporary speaker does is inheritance and inhabitation of a grip established for her. The restatement objection misses the direction of derivation; the descriptivism objection misplaces reference in the speaker's beliefs; the categorial-feature objection mistakes a describable practice for a relabelled conclusion; the perception objection collapses a correctable word-use into an uncorrectable link. Once reference-fixing is put back at the origin, where the account places it, each objection loses its grip. None of this shows the account is true. It shows the account is not any of the four things it is most likely to be mistaken for.

7. Conclusion

7.1. What Has Been Argued

The dissertation began from a gap in Kripke. The semantic results of *Naming and Necessity* tell us what rigid designators do — hold their referent fixed across every world in which it exists — and Kripke argues for those results on grounds that owe nothing to the necessity-of-identity formula. What the results do not tell us is what cognitive operation a speaker performs such that her words behave as the semantics describes. Kripke saw that reference to a particular cannot be secured by formal operations alone, which is why he appealed to baptism and the causal chain; but he offered that appeal as "a better picture," not a theory, and left the operation it pictures unanalyzed. The dissertation set out to analyze it.

The analysis ran as follows. Reading the necessity-of-identity formula closely (Chapter 2) showed that the word "object" in its natural-language gloss spans three things — variable, encountered particular, named individual — and that the formula's validity holds for the first while its philosophical payload concerns the third. The move between the levels is not made by the formula; it is where a cognitive mechanism is needed and not supplied. Tracing the Russell–Kripke line (Chapter 3) showed that descriptions cannot supply the missing rigidity from within, and that Russell had located the right mechanism — direct, non-descriptive coordination — only to confine it to incorrigible sense-data by tying directness to infallibility. Prising those two properties apart is what frees the mechanism for ordinary objects. Chapter 4 named the mechanism: the x^o/x^c functional-mode distinction, with R as the coordination between a word and what it is used about, and *local rigidity* as the rigidity an x^o-coordination has constitutively, for as long as the coordination lasts. The same distinction re-read the de re/de dicto ambiguity and dissolved the air of paradox around the necessary a posteriori, by locating necessity at x^o and a-posteriority at x^c. Chapter 5 gave the constructive account: baptism *lifts* a live x^o-coordination into a portable name-concept N^c, the causal chain is the practice that sustains N^c across generations, and contemporary speakers engage absent bearers through revisable attributions that ride on a reference they do not constitute. Chapter 6 defended this against the objections it most invites.

The single claim the dissertation has tried to establish is that Kripkean rigidity is not primitive. It is local rigidity made portable. The rigidity that names have is the rigidity that demonstrative coordination has, carried past the encounter by a social practice that did not create it but only relocated its locus. This is a metasemantic claim, not a rival semantics: it leaves Kripke's model untouched and explains how speakers come to inhabit it.

7.2. What Remains Open

The thesis was deliberately narrow, and its scope leaves several questions standing. I mark them honestly, since an account is more useful with its edges visible.

General-term rigidity. The \$ x^o/ x^c\$ distinction applies symmetrically to common nouns and names (§5.1), and the account therefore has the materials to ask whether and how "water" or "gold" are rigid. But the question of general-term rigidity has its own large literature and its own complications — the role of scientific theorizing in fixing kind-reference, the metaphysics of natural kinds — and the dissertation took no stand on them (Ch. 4, "The Necessary *A Posteriori*," n.19). Whether local rigidity scales to kinds the way it scales to individuals is the most natural extension of the account and the first thing I would take up next.

Empty and fictional names. The account anchors reference in a baptismal R-grounding with a real individual. Names with no such origin — "Pegasus," "Vulcan," names from fiction — therefore fall outside the mechanism as stated. This is not obviously a defect: it predicts that empty names behave differently from genuine ones, which they do. But it owes an account of *what* the practice is doing when it transmits a name with no baptismal anchor, and the conjecture that fictional names involve a pretended or bracketed grounding (a baptism within a pretense) is only a conjecture here.

Mathematical reference. Mathematical objects are rigid — "the number eight" picks out the same number across worlds — but there is no perceptual encounter to ground an \$ x^{o}\$-coordination with a number. The account treats mathematical rigidity as A-grounded (§2.2): fixed within the linguistic-formal system, not by reaching outside it. Whether that is the whole story, or whether mathematical practice involves something analogous to baptism, is left open.

The diagnostic application to Russell, Quine, and Lewis. The introduction suggested that a shared over-reliance on formal operations runs through Russell's sense-data, Quine's bound variables, and Lewis's plurality of worlds, and that the A-/R-grounding distinction diagnoses it. The dissertation used Russell but reserved the full historical treatment for separate work (Ch. 1, "Methodology and Scope"). That treatment would test whether the distinction is as diagnostically general as the introduction proposed, or whether it illuminates Russell and overreaches on the other two.

The developmental and anthropological question. Chapter 6 located the persistence of reference in a community practice of origin-tracking correction, and conceded that *why* human communities maintain such a practice for names is a further question the account gestures at but does not answer. A full genealogy of the naming practice — how the lifting of local rigidity becomes a stable social institution — would connect this work to developmental psychology and to the study of linguistic norms, and would test the claim that the practice exists because carrying a fixed grip past the encounter is useful.

7.3. The Shape of the Result

What the dissertation offers is small and specific: not a theory of reference, but the closing of one gap in an existing theory. Kripke described a practice and declined to analyze the operation it rests on. The operation, I have argued, is one we can watch in its simplest form whenever a speaker points at a bird and says something about how it might have been — the holding-fixed of a particular across everything imagined about it. Names are that holding-fixed, made to survive the absence of the thing held. The cognitive precondition Kripke presupposed is the same one Russell found and abandoned; what the account does is take it back from the place Russell left it and follow it to where Kripke needed it.

8. Bibliography

8.1. Primary Literature

(Texts the dissertation directly analyzes or argues against)

Kaplan, D. (1968). "Quantifying In." *Synthese*, 19(1–2), 178–214.

Kaplan, D. (1989). "Demonstratives." In J. Almog, J. Perry & H. Wettstein (eds.), *Themes from Kaplan*. Oxford: Oxford University Press, pp. 481–563.

Kripke, S. A. (1971). "Identity and Necessity." In M. K. Munitz (ed.), *Identity and Individuation*. New York: New York University Press, pp. 135–164.

Kripke, S. A. (1980). *Naming and Necessity*. Cambridge, MA: Harvard University Press.

Quine, W. V. O. (1953). "Reference and Modality." In *From a Logical Point of View*. Cambridge, MA: Harvard University Press, pp. 139–159.

Russell, B. (1905). "On Denoting." *Mind*, 14(56), 479–493.

Russell, B. (1918). "The Philosophy of Logical Atomism." *The Monist*, 28–29. [Repr. in R. C. Marsh (ed.), *Logic and Knowledge*. London: Allen & Unwin, 1956, pp. 177–281.]

Wittgenstein, L. (1922). *Tractatus Logico-Philosophicus*. Trans. C. K. Ogden. London: Routledge & Kegan Paul.

Wittgenstein, L. (1953). *Philosophical Investigations*. Trans. G. E. M. Anscombe. Oxford: Blackwell.

8.2. Secondary Literature

Devitt, M. (1981). *Designation*. New York: Columbia University Press.

Dickie, I. (2015). *Fixing Reference*. Oxford: Oxford University Press.

Evans, G. (1982). *The Varieties of Reference*. Oxford: Oxford University Press.

LaPorte, J. (2022). "Rigid Designators." In E. N. Zalta (ed.), *The Stanford Encyclopedia of Philosophy* (Spring 2022 ed.). https://plato.stanford.edu/entries/rigid-designators/

Putnam, H. (1975). "The Meaning of 'Meaning'." In *Mind, Language and Reality: Philosophical Papers, Vol. 2*. Cambridge: Cambridge University Press, pp. 215–271. [Also *Minnesota Studies in the Philosophy of Science*, 7, 131–193.]

Recanati, F. (2012). *Mental Files*. Oxford: Oxford University Press.

Salmon, N. (1981). *Reference and Essence*. Princeton: Princeton University Press.

Smullyan, A. F. (1948). "Modality and Description." *Journal of Symbolic Logic*, 13(1), 31–37.

Soames, S. (2002). *Beyond Rigidity: The Unfinished Semantic Agenda of* Naming and Necessity. New York: Oxford University Press.

8.3. Author's Own Work

Cojocariu, F. (2025). "Immediate and Mediated Existence: An Inquiry into Different Uses of 'Exists'." Unpublished manuscript.

Cojocariu, F. (2026). "Pattern-Recognition Unity: A Framework Specification." Preprint. https://dx.doi.org/10.2139/ssrn.6285878

[1] Cojocariu F. Pattern-Recognition Unity (PRU): A Framework Specification 2026. https://doi.org/10.2139/ssrn.6285878.